
Automated Circuit Interpretation via Probe Prompting: Concept-Aligned Supernodes and a Matched-Control Causal-Validation Harness for Attribution Graphs

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Abstract

Manual interpretation of attribution graphs is slow (approximately 2 hours per prompt) and yields inconsistent taxonomies. We introduce *probe prompting*, a transparent rule-based pipeline that converts a graph into concept-aligned supernodes via Cross-Prompt Activation Signatures (CPAS). We define an *operationally useful* label as one that satisfies two conditions: (i) **Specificity**—intervention via the labeled supernode redirects output toward the specific concept, and (ii) **Matched-Control**—intervention via a structurally matched random feature set (same count, same per-layer histogram, sampled outside concept-matching supernodes) does not produce the same effect. Across 5 domains and 42,184 entity-swap interventions, labeled supernodes satisfy both conditions in 4 of 5 domains (USA 67.3 vs 0.2% hit; Books 63.8 vs 0.0%), with Sounds (6 entities, structural confounds) failing Condition 2 (matched-control vsMax gap +0.14) and reported as a directional negative. The harness surfaces three regularities: input-field features *interfere* with steering (less-is-more, +14–33 pp), the canonical $M=20$ default is *over-steering* (winning M is bimodal at ~ 2.4 and ~ 6.9), and graph scaffold compatibility is a perfect ordinal predictor of domain-level success. Code, datasets, and an interactive demo are released anonymously.

1. Introduction: An Operational Test for Concept-Aligned Supernodes

The interpretation bottleneck. Attribution graphs scale faster than humans can read them. A single Gemma-2-

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2B-it graph for a short factual prompt routinely contains 600–5,000 cross-layer transcoder (CLT) features and edges between them (Ameisen et al., 2025; Lindsey et al., 2025a). Manual interpretation by an experienced circuit tracer is reported to take approximately two hours per prompt (Neuronpedia, 2025). Existing autointerp pipelines (Bills et al., 2023; Templeton et al., 2024; Bricken et al., 2023) label individual features from corpus activations, but they do not produce *circuit-level* groupings calibrated to a prompt’s target logit, and they are not validated by intervention. As mechanistic interpretability moves from single circuits to large-scale catalogs (Hanna & Ameisen, 2026; Marks et al., 2025), an automated, behavior-grounded, and *causally validated* grouping is needed.

Approach: probe prompting. We propose *probe prompting*, a transparent rule-based pipeline that converts an attribution graph into compact subgraphs of *concept-aligned supernodes*. For each candidate feature in the graph, we project a small set of concept-targeted prompts (the “lights”) and record the per-feature Cross-Prompt Activation Signatures (CPAS) (the “shadows”). Threshold rules then assign one of four functional roles—Semantic-Dictionary, Semantic-Concept, Relationship, Say-X—and same-role same-name features are grouped into supernodes. The pipeline is deterministic, auditable, and editable; learned classifiers were rejected because the grouping itself is the object of interpretability research and must be inspectable (Rudin, 2019).

An operational test, not a similarity test. Behavioral coherence relative to a geometric clustering baseline (cosine, layer-adjacency) is structurally biased: any grouping derived from a behavior-grounded signature outperforms a purely geometric grouping on behavior-grounded metrics, near-tautologically. Heap et al. (2025) sharpen this concern by showing that randomly initialized transformers can score competitively on common interpretability metrics, undermining metrics that are not anchored to causal effect. We therefore validate supernodes by *intervention against structurally matched random controls*, in line with recent causal-circuit work on transcoders (Hanna & Ameisen,

2026; Geiger et al., 2025). We deliberately do *not* claim full mechanistic identification; our supernode labels are behavioral abstractions with falsifiable causal consequences.

Two falsifiable claims. C1 (operationally useful labels).

A supernode label is *operationally useful* for entity e if intervening on the supernode redirects output toward e (**Specificity**, Condition 1) and a structurally matched random feature set, sampled outside concept-matching supernodes, does not (**Matched-Control**, Condition 2). *Falsification*: either Specificity hit-rate fails to exceed Matched-Control hit-rate by ≥ 5 pp on a per-pair basis, or labeled vsMax is within 1.0 logits of Matched-Control vsMax, in any of the four strong domains (USA, Books, Products, Paintings).

C2 (mechanistic structure surfaced by the pipeline). The harness reveals four directional sub-claims, each with quantitative thresholds: (i) a backbone-and-specialization layer gradient (early/late Jaccard ratio > 1.4); (ii) a less-is-more / field-interference effect (intermediate+answer beats all-3-fields by ≥ 10 pp in $\geq 3/5$ domains); (iii) over-steering at the canonical $M=20$ default (adaptive M -search rescues $\geq 20\%$ of misses; high- M and top- k controls rescue zero); (iv) graph scaffold compatibility as a perfect ordinal predictor of domain-level hit rate across the four strong domains.

What we explicitly do not claim. Following the three-levels framing of Geiger et al. (2025), we claim Level 1 (operationally useful labels) and Level 2 (downstream causal effects with matched controls) but not Level 3 (full mechanistic identification of latent variables). We do not propose a new attribution method or clustering algorithm. The setting is Gemma-2-2B-it with the public CLT-HP dictionary on single-hop factual recall; cross-architecture and chain-of-thought generalization are open.

Contributions.

1. **Pipeline.** A transparent, rule-based pipeline that converts attribution graphs into concept-aligned supernodes from cross-prompt activation signatures (Section 4).
2. **Causal-validation harness.** A matched-random-control swap protocol with field-additivity decomposition and adaptive M -search, applicable to any feature-grouping method (Section 4, Section 5).
3. **Substantive findings.** Less-is-more, over-steering at $M=20$, and scaffold-as-predictor (Section 6).
4. **Open release.** A 42,184-run dataset, anonymous code, and an interactive demo (Appendix A).

2. Definitions: An Operationally Useful Supernode Label

We follow Hanna & Ameisen (2026) in stating definitions explicitly so that the falsifiability conditions are exposed.

Setting. Let $\mathcal{M}$ be a transformer (Gemma-2-2B-it), $\mathcal{D}$ a CLT-HP feature dictionary (Ameisen et al., 2025), and $G_p = (V_p, E_p)$ the attribution graph for prompt p with target logit y . Each node $f \in V_p$ is a (feature, position) pair with influence $\text{infl}(f) \in \mathbb{R}_{\geq 0}$; the cumulative-influence threshold τ selects a feature universe $V_p^\tau \subseteq V_p$.

Definition 2.1 (Supernode). A *supernode* $s = (\rho, \nu, F)$ is a triple of a functional role $\rho \in \{\text{SEM-DICT}, \text{SEM-CONC}, \text{REL}, \text{SAY-X}\}$, a string label ν , and a feature set $F \subseteq V_p^\tau$ all sharing role ρ and label ν . Each feature belongs to at most one supernode (Section 4).

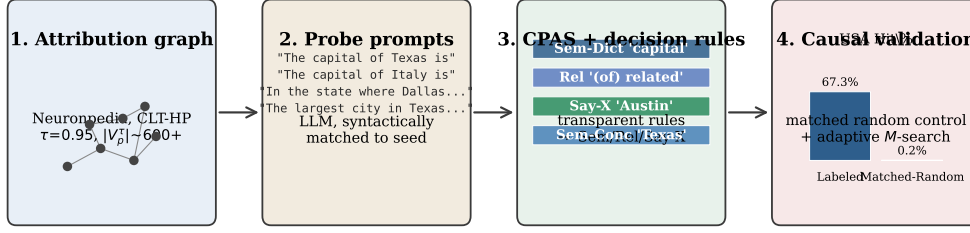
Definition 2.2 (Concept-aligned supernode). A supernode $s = (\rho, \nu, F)$ is *aligned to concept* c if (i) ν matches c under the substring-aware matcher of Section 4, or (ii) $\rho = \text{SAY-X}$ and the discovered target token of s matches c . Each entity e in a domain induces a set of concepts $C(e)$ across its semantic fields (input, intermediate, answer); the corresponding supernodes are $S(e) = \{s : s \text{ aligned to some } c \in C(e)\}$.

Definition 2.3 (Additive entity-swap intervention). For a swap from source entity e_A to target entity e_B on prompt p , the intervention $\mathcal{I}(e_A \rightarrow e_B; M_a, M_b)$ ablates each feature $f \in \bigcup_{s \in S(e_A)} F$ by additive decoder injection with multiplier M_a (default -2) and amplifies each $f \in \bigcup_{s \in S(e_B)} F$ with M_b (default 20). Attention is *not* frozen during intervention (FREEZE_ATTENTION=false), so the model can route around the perturbation (Ameisen et al., 2025).

Definition 2.4 (Hit and vsMax). Let $g(p, \mathcal{I})$ be the generated continuation under intervention $\mathcal{I}$ at temperature 0.3, $n_{\text{tokens}}=10$, frequency penalty 2.0, seed 42. A swap is a **Hit** for target e_B iff the first-subword token of e_B ’s answer appears in g . The metric **vsMax** is $\max_{t \leq 10} (\ell_{e_B}(t) - \max_{e' \in \mathcal{A} \setminus \{e_B\}} \ell_{e'}(t))$, where $\ell_e(t)$ is the logit of e ’s first-subword token at trajectory position t and $\mathcal{A}$ is the domain answer set.

Definition 2.5 (Matched random control). The *matched random control* for swap pair (e_A, e_B) is an intervention $\mathcal{I}^{\text{rand}}$ identical to $\mathcal{I}(e_A \rightarrow e_B)$ in feature count and per-layer histogram, with features sampled uniformly at random from $V_p^\tau \setminus \bigcup_{e \in \mathcal{E}} \bigcup_{s \in S(e)} F$ (i.e., outside *every* concept-aligned supernode in the domain). Three deterministic replicates are drawn per pair, seeded by sha256(run_seed : pair_id : rep : mode).

This control directly responds to the Heap et al. (2025) critique: any structural property shared by “random” features (count, layer distribution, eligibility for additive injection)



Operationally useful label (Def. 6): Specificity (intervention redirects to e) AND Matched-Control (structurally matched random feature set does not).

Figure 1. Pipeline overview. Four stages: (1) attribution graph generation (Neuronpedia, CLT-HP features, $\tau=0.95$); (2) probe prompts (LLM-generated, syntactically matched to seed); (3) Cross-Prompt Activation Signatures and transparent decision rules $\rightarrow$ typed supernodes; (4) **causal-validation** via additive feature-swap interventions with structurally matched random controls (same feature count, same per-layer histogram, sampled outside concept-matching supernodes). The unit of analysis is the supernode, validated operationally by Conditions 1 and 2 of Definition 2.6.

is held constant; only the *concept alignment* of the chosen features changes.

Definition 2.6 (Operationally useful label). The supernode label ν for entity e is *operationally useful* on prompt family p iff both conditions hold over all swap pairs (e_A, e_B) in the domain:

- **Condition 1 (Specificity).** The Hit-rate of $\mathcal{I}(e_A \rightarrow e_B)$ exceeds 5 pp on a per-pair basis, and labeled vsMax is positive on average.
- **Condition 2 (Matched-Control).** Across replicates, $\mathcal{I}^{\text{rand}}$ achieves Hit-rate at least 5 pp lower than the labeled intervention and labeled vsMax exceeds matched-control vsMax by at least 1.0 logit.

A label fails to be operationally useful if either condition fails. *This is the falsification criterion for C1.*

Why two conditions, not one. Condition 1 alone permits a label to “work” for trivial reasons (e.g., the intervention magnitude pushes the logit distribution arbitrarily). Condition 2 forces the redirection to be *specific to concept alignment*. Hanna & Ameisen (2026) adopt an analogous two-condition structure for latent planning—a candidate planning feature must (i) precede the planned token and (ii) causally affect the planned token under intervention.

Three levels of interpretive claim. We follow the framing of Geiger et al. (2025): Level 1 (operationally useful labels), Level 2 (downstream causal effects, validated against controls), Level 3 (full mechanistic identification of latent variables). Definition 2.6 formalizes Levels 1–2; we explicitly disclaim Level 3 (Section 7).

3. Related Work: Attribution Graphs, Autointerp, and Causal Validation

Attribution graphs and replacement models. Attribution graphs operationalize feature $\rightarrow$ logit pathways in a local replacement model that linearizes the residual stream through CLT or SAE features and freezes attention (Ameisen et al., 2025; Lindsey et al., 2025a). The Neuronpedia platform exposes these graphs and per-feature cards at scale (Neuronpedia, 2025; Lindsey et al., 2025b), and circuit-tracer provides the open-source reference implementation (Safety Research contributors, 2025). Our work operates entirely *downstream* of attribution: we do not propose a new attribution method, but we do produce supernodes that the same Neuronpedia tooling can render and pin.

Sparse features and dictionary learning. Sparse Autoencoders and Cross-Layer Transcoders expose sparse, often monosemantic features that make replacement-model analyses tractable (Bricken et al., 2023; Templeton et al., 2024; Transformer Circuits Team, 2024; Marks et al., 2025). We treat the CLT-HP dictionary as fixed and study how to *group* its features into circuit-level units.

Feature labeling and autointerp. Automated interpretation of individual features by LLM scoring (Bills et al., 2023; Templeton et al., 2024) produces per-feature labels from corpus activations but does not yield circuit-level groupings calibrated to a target logit, and is not validated by intervention. We take the complementary route: prompt-conditioned, circuit-level grouping with explicit causal validation.

Concurrent work on causal circuits with transcoders. Hanna & Ameisen (2026) use cross-layer transcoder fea-

ture circuits to study whether language models latently plan rhyming words, structuring their analysis around two falsifiability conditions (predict-token-before-emission, causally-affect-token). Our two-condition framing of Definition 2.6 (Specificity, Matched-Control) is directly inspired by their template, applied to the more general question of supernode label utility. Marks et al. (2025) apply sparse-feature circuit analysis to subject–verb agreement; we contribute a domain-general harness with structurally matched random controls.

Steering with sparse features. Single-feature steering with high multipliers (Anthropic’s $M \sim 20$ paradigm) has been used to demonstrate concept-specific behavior (Templeton et al., 2024; Kharlapenko et al., 2024). We generalize to multi-feature concept-aligned swaps and find empirically that the canonical $M=20$ default is over-steering in our setting (Section 6), with the optimal M bimodal at ~ 2.4 and ~ 6.9 .

Causal interpretability and matched random controls. Geiger et al. (2025) formalize three levels of interpretive claim and emphasize that *distinguishing* levels requires intervention. Heap et al. (2025) show that randomly initialized transformers can score competitively on common interpretability metrics, undermining metrics that lack causal anchoring. Our matched-random-control protocol (Definition 2.5) is designed to address this critique directly: any structural property a random feature set shares with the labeled one (count, layer histogram, eligibility) is held constant; only the concept alignment changes.

Why we do not include a geometric-clustering baseline. A grouping derived from a behavior-grounded signature outperforms a purely geometric grouping (cosine, layer-adjacency, k-means, HDBSCAN) on behavior-grounded metrics by construction, so the comparison is structurally biased in our favor; and there is no canonical “feature clustering” baseline in the SAE/CLT literature to anchor the comparison. We instead validate via intervention against matched random controls (Definition 2.5), which is robust to the bias and *applicable to any grouping method*—including future learned ones. We discuss this choice as a limitation in Section 7.

4. Method: Probe Prompting and a Matched-Control Causal-Validation Harness

4.1. Setting and attribution-graph generation

We use Gemma-2-2B-it with the public CLT-HP dictionary (~ 2.5 M features over 26 layers; Ameisen et al., 2025). Attribution graphs are generated via the Neuronpedia API

with `nodeThreshold=0.8`, `edgeThreshold=0.85`, `maxFeatureNodes=5000`. The feature universe V_p^τ is selected by cumulative-influence threshold $\tau=0.95$ for batch experiments. The replacement model freezes attention during graph computation; downstream interventions (Definition 2.3) run with attention free, making positive results harder than under frozen attention (Ameisen et al., 2025).

4.2. Cross-Prompt Activation Signatures (CPAS)

For each (feature, probe) pair we record raw measures (peak token, peak position, activation density, sparsity ratio, robust z , cosine to seed). Aggregating across probes yields the CPAS used by the decision tree below: `peak_consistency_main`, `n_distinct_peaks`, `func_vs_sem_pct`, `share_F`, `sparsity_median`, `conf_S`. Formal definitions are deferred to Appendix B.

4.3. Functional vs. semantic tokens; target-token mapping

Tokens are labeled *functional* (copulas, articles, prepositions, conjunctions, relative pronouns, auxiliaries; ~ 87 English tokens) or *semantic* (content-bearing). When a feature peaks on a functional token, a ± 7 -token directional search identifies the nearest semantic target (forward for `is/are/articles/most` prepositions; backward for `of/possessive 's`; bidirectional for conjunctions/punctuation). Full vocabulary, edge cases, and a documented French-prompt failure mode are in Appendix C.

4.4. Transparent classification rules

Feature classification uses the strict-priority decision tree below; the first matching rule wins. Pseudocode and threshold derivations are in Appendix B.

- SEM-DICT: `peak_consistency_main` ≥ 0.80 AND `n_distinct_peaks` ≤ 1 .
- SAY-X: `func_vs_sem_pct` ≥ 50 AND `conf_F` ≥ 0.90 AND `layer` ≥ 7 .
- REL: `sparsity_median` < 0.45 .
- SEM-CONC: `layer` ≤ 3 OR `conf_S` ≥ 0.50 OR `func_vs_sem_pct` < 50 .
- REVIEW: otherwise (manual inspection, excluded from analysis).

We chose rules rather than a learned classifier because the grouping itself is the object of interpretability research. Rules give traceability (every assignment maps to specific threshold crossings), debuggability, editability, and communicability—properties a learned grouper would lose.

Table 1. Pipeline output: compression with high coverage. *Repl.* = fraction of end-to-end influence routed via features; *Comp.* = fraction of incoming edge influence explained. Concept-aligned subgraphs preserve Completeness (0.83 vs. 0.90) while compressing $\sim 600+$ raw features into 30–50 supernodes (*Repl.* drops 0.70 $\rightarrow$ 0.54). Descriptive only.

Prompt	G-Repl.	S-Repl.	G-Comp.	S-Comp.
Austin (Dallas)	0.72	0.57	0.90	0.83
Oakland (swap)	0.70	0.57	0.90	0.83
Michael Jordan	0.69	0.49	0.90	0.81
Small opposite	0.73	0.62	0.91	0.86
Muscle diaphragm	0.63	0.45	0.87	0.79
Mean	0.70	0.54	0.90	0.83

4.5. Supernode formation and naming

Same-rule same-name features form a supernode (Definition 2.1). Cross-prompt stability is enforced ($\geq 60\%$ of probes); features failing this are marked *ungrouped* (5–10% of candidates) and excluded. Naming policy: strongest semantic peak for SEM-DICT/SEM-CONC; discovered target token for SAY-X; aggregated activation over an extended vocabulary for REL (format “(token) related”). A configurable blacklist filters uninformative names.

4.6. Pipeline output: compression with high coverage

The pipeline output is a property of the method, not a claim being tested: concept-aligned subgraphs compress 600+ raw features into 30–50 named supernodes per circuit while preserving Neuronpedia *Completeness* (mean 0.83 over five circuits) and accepting a *Replacement* drop to 0.54 (Table 1). The trade-off is intentional: low-influence nodes are treated as error to maximize legibility. We report this here so the reader knows the unit of analysis (supernodes, not raw features) before Section 6 introduces interventions.

4.7. Causal-validation harness

Intervention mechanism. Additive delta injection via CLT decoder vectors (Definition 2.3). For each intervened feature, $v_{\text{new}} = M \cdot v_{\text{orig}}$ scales the decoder vectors and adds the result to the residual stream at downstream layers. Defaults: $M_{\text{ablate}} = -2$, $M_{\text{amplify}} = 20$. *Attention is not frozen* during intervention; positive results are stronger evidence than under frozen-attention patching where some direct effects are forced by construction (Ameisen et al., 2025).

Matched random control (Definition 2.5). This is the central methodological move and replaces the geometric-baseline comparison. The control fixes feature count and per-layer histogram and excludes *all* concept-aligned supernodes from the sampling pool, so the only varying property is concept alignment. Three deterministic replicates per pair

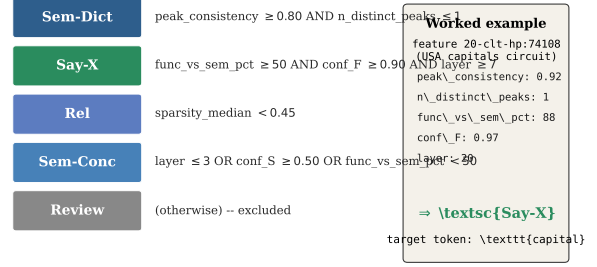


Figure 2. Transparent decision tree (Section 4.4). Strict priority ordering; the first matching rule wins. The worked example traces feature 20-clt-hp:74108 from the USA capitals circuit (CPAS in Appendix M), classified as SAY-X with target token *capital*.

give Hit-rate and vsMax distributions over random feature sets. The protocol is grouping-method-agnostic and can be re-run on any competing grouping.

Field-additivity decomposition. Each domain has 3 semantic fields (input, intermediate, answer; Table 2). For each pair, 7 variants are run: 3 single-field, 3 two-field, 1 all-three. Each selected field drives both ablation and amplification; this isolates which fields carry the causal signal.

Adaptive M -search. Two-phase post-hoc search for M_{amplify} on missed pairs. Phase 1: 6-point geometric probe on $[0.1, 20]$, stopping at the first hit. Phase 2 (if no Phase 1 hit): KL-transition binary refinement around the steepest KL transition, 6 steps. Applied to labeled and field-additivity runs (1,210 + 867 eligible pairs; full pseudocode in Appendix G).

4.8. Metrics

Primary (per Definition 2.4): **Hit%**, **vsMax**, **Target Recovery** (target logit beats its own unsteered baseline at any trajectory position). **Supporting:** **Sup%** (source absent from output), **RkGrp** (best target rank within the domain answer set), **Flip%** (target–source rank crossover), **CtrlS** (mean absolute logit shift of control tokens), **KL(baseline||steered)** at position 0. We deliberately do *not* use gap-closure as a primary metric (regime-dependent and confounded by baseline gap; see Appendix D). We report both position-0 and best-over-trajectory variants; the two diverge in informative ways across domains (Appendix H).

5. Experimental Design: Five Domains, 42,184 Swap Runs

5.1. Domains and seed prompts

We evaluate on five domains spanning single-hop factual recall in different relational structures and answer geome-

tries (Table 2). The first four are the *strong* domains used for cross-domain inference; *Sounds* (6 entities, 12/30 pairs share answers, $K=6$) is reported separately due to known structural confounds (Section 6.3, Section 7).

5.2. Experimental matrix

Each swap pair is run under: (a) labeled (3-field) baseline at $M=20$; (b) matched random control (3 replicates per pair); (c) field-additivity (7 variants per pair); (d) adaptive M -search on labeled and on the best field-add variant per missed pair. The total is **42,184 swap runs** across all conditions, domains, and M values (per-condition counts in Table 2).

5.3. Pre-registered evaluation choices

Before analysis we fixed: (i) contrast group $\mathcal{A}$ = all other dataset answers; (ii) first-subword resolution for multi-token answers; (iii) explicit `answer_field` decoupling intervention from measurement; (iv) generation parameters ($T=0.3$, $n=10$, frequency-penalty 2.0, seed 42); (v) sha256-seeded matched-control replicates; (vi) pair-level definition of Hit (any of 3 random replicates is a hit counts as a random-hit for that pair). Ranges, thresholds, and seeds were frozen prior to running the 42,184-run sweep.

6. Results

We organize results by claim. Each subsection ends with an explicit **Falsification verdict** relative to the thresholds in Sections 1 and 2.

6.1. C1: Concept-aligned supernodes carry entity-specific causal leverage that matched random controls do not

Headline (Table 3). With per-pair best field-add variant and adaptive M -search, labeled hit-rates dwarf matched-random controls in 4 of 5 domains:

Specificity decomposition. Random controls achieve *equal or higher* suppression than labeled in 3 of 5 domains (Books 75 vs 69%; Products 87 vs 65%; Paintings 74 vs 34%; Appendix E). Suppression is generic; *only targeting is specific*. This invalidates raw Sup% as a label-quality metric and motivates Hit%/vsMax/Target-Recovery as the discriminating signals.

Within-disruption-regime evidence (Regime C). Classifying every swap by what happens to target and source logits at position 0 yields four regimes (A: clean redirect; C: differential disruption with flip; D: generic disruption; E: pure suppression; Appendix L). Random controls concentrate in regime D (42–45% of pairs); the labeled best

variant concentrates in regime A. *Within* regime C, where both logits are disrupted but the target overtakes, labeled drives target above its unsteered baseline 92% of the time vs. 29% for random in USA (Figure 3b). This is the cleanest within-regime signal that “the model’s circuits responded to the target concept” rather than “everything got disrupted.”

Statistical reporting. Bootstrap 95% CIs on hit-rate gaps and permutation tests on overlap statistics are reported in Appendices E and J. With 2,450 USA pairs and 240 Books pairs, the observed Hit-rate gaps (67.1 and 63.8 pp) lie far outside any plausible null.

Falsification verdict.

- **C1 supported** in USA, Books, Products, Paintings (Specificity gap ≥ 5 pp and vsMax gap ≥ 1.0 logit; Definition 2.6).
- **C1 not supported in Sounds** (vsMax gap +0.14, Hit-rate gap +16.7 pp on tiny $N=30$). Reported as a directional negative result in Section 6.3.

6.2. C2: The harness surfaces four mechanistic regularities

6.2.1. C2.1: LESS IS MORE—INPUT-FIELD FEATURES INTERFERE WITH STEERING

In 4 of 5 domains, the intermediate+answer subset beats the all-3-field intervention by 9–33 pp (Table 4):

- USA state+capital 38.8% vs. all-3 24.7% (+14.1 pp);
- Books book+author 37.1% vs. all-3 3.8% (+33.3 pp);
- Products company+founder 24.2% vs. all-3 15.2% (+9.0 pp);
- Sounds sound+animal 20.0% vs. all-3 0.0% (+20.0 pp);
- Paintings: weak baseline, weak gain (+3.4 pp).

Interpretation. Input-field features encode what the model *reads*, not what it should *produce*. Including them activates competing circuits that dilute or interfere with the answer signal. Cross-domain single-field aggregates confirm the role gradient: intermediate (Hit 14.6%) > answer (8.8%) > input (6.1%) (Appendix F).

6.2.2. C2.2: DEFAULT $M=20$ IS OVER-STEERING; THE WINNING M DISTRIBUTION IS BIMODAL

Adaptive M -search rescues 452 missed pairs across all conditions. The winning- M distribution is bimodal at $M \sim 2.4$ (47% of labeled hits) and $M \sim 6.9$ (40%) (Figure 4). Two control experiments confirm the mechanism rather than the tuning:

Table 2. Five evaluation domains. Each row gives the seed-prompt template, the three semantic fields (input / intermediate / answer), the answer-set size K , the number of entities and non-identity swap pairs, and the total run count across all conditions (labeled, matched-random $\times 3$, 7 field-additivity variants, adaptive- M rescue). The four *strong* domains (USA, Books, Products, Paintings) are used for cross-domain inference; *Sounds* is reported separately as a directional negative (Section 6.3). Full per-condition counts in Appendices E to G.

Domain	Seed prompt template	Fields (in / int / ans)	K	Entities	Pairs	Total runs
USA	“capital of state containing X ”	city / state / capital	50	50	2,450	$\approx 28,400$
Books	“character X from book by”	character / book / author	25	16	240	$\approx 2,800$
Products	“product X from company by”	product / company / founder	25	12	132	$\approx 1,700$
Paintings	“painting X by painter”	painting / painter / first_name	20	10	90	$\approx 1,200$
Sounds	“sound X from animal of color”	sound / animal / color	6	6	30	≈ 400
Total						42,184

Table 3. Headline (C1): labeled supernodes vs. matched random controls. For each domain we report (a) the per-pair best field-add variant with adaptive M -search Hit% (a comprehensive operational measure), and (b) the labeled vs. matched-random vsMax discriminator at the full-labeled all-fields $M=20$ baseline (the cleanest specificity signal; full per-condition tables in Appendix E). Hit% counts pairs whose first generated subword matches the target answer; vsMax is the best logit margin against the strongest competing dataset answer. C1 (Definition 2.6) is supported when both Hit-rate gap ≥ 5 pp and vsMax gap ≥ 1.0 logit. Sounds fails Condition 2 (vsMax gap $+0.14$) and is reported as a directional negative (Section 6.3).

Domain	Hit% (best-FA, M -opt)			vsMax (full lab. vs. rand., $M=20$)			Verdict
	Lab.	Rand.	Δ	Lab.	Rand.	Δ	
USA	67.3	0.2	+67.1	+2.86	-2.31	+5.17	C1 supported
Books	63.8	0.0	+63.8	+5.98	-0.15	+6.13	C1 supported
Products	27.6	1.1	+26.5	+3.46	+0.23	+3.23	C1 supported
Paintings	21.8	0.0	+21.8	+1.55	-0.03	+1.58	C1 supported
Sounds	20.0	3.3	+16.7	+3.28	+3.14	+0.14	<i>Negative</i> (Section 6.3)

Table 4. Less-is-more (C2.1): smaller field subsets beat the all-three-field intervention. Per-domain best subset vs. the all-3-field labeled baseline at $M=20$. In 4/5 domains the intermediate field carries the strongest signal; pairing it with the answer field yields a 9–33 pp gain. Per-variant tables in Appendix F.

Domain	Best subset	Best Hit%	All-3 Hit%	Δ
USA	state + capital	38.8	24.7	+14.1
Books	book + author	37.1	3.8	+33.3
Products	company + founder	24.2	15.2	+9.0
Sounds	sound + animal	20.0	0.0	+20.0
Paintings	first_name (1-flt)	6.7	3.3	+3.4

- **High- M rescue is null.** Testing $M \in \{50, 100, 200\}$ on near-miss pairs (80 runs across 3 domains) produced *0 new hits*. KL saturates above $M \sim 50$ while output quality continues to worsen.
- **Top- k rescue is null.** Testing top- k features by graph influence ($k \in \{1, 3, 5, 10\}$) at $M \in [20, 200]$ on Products produced *0/96 hits*. Only the full feature set ($k=67$) achieves hits.

Interpretation. The steering signal is *distributed*, not concentrated. Multi-feature amplification at high M produces distributed noise that overwhelms the target signal. This contrasts structurally with the single-feature paradigm in

which high M is a targeted directional push (Templeton et al., 2024).

6.2.3. C2.3: BACKBONE-AND-SPECIALIZATION LAYER GRADIENT

Cross-prompt scalable analysis on 1,607 intra-domain pairs across 5 domains (103 entities; Appendix J): activation stability $> 90\%$ in all domains; Jaccard overlap 0.29–0.62 (significantly above chance, $p < 0.001$ via permutation at multiple pool sizes). Early/late Jaccard ratio is $1.4\text{--}3.0 \times$ in 4 of 5 domains: early layers shared backbone, late layers entity-specific.

6.2.4. C2.4: SCAFFOLD INFLUENCE IS A PERFECT ORDINAL PREDICTOR OF DOMAIN-LEVEL SUCCESS

Graph scaffold compatibility—the fraction of total graph influence carried by features with *identical* supernode assignment across two entity graphs—perfectly orders the four strong domains by Hit% at the all-3-field labeled baseline (Spearman $\rho=1.0$, $N=4$; Appendix K): USA 53.0% $\rightarrow$ 24.7%; Products 42.2% $\rightarrow$ 15.2%; Paintings 35.9% $\rightarrow$ 4.4%; Books 25.3% $\rightarrow$ 3.8%. The late-layer ($\geq L_{16}$) scaffold gradient is even steeper (USA 16.3% vs Books

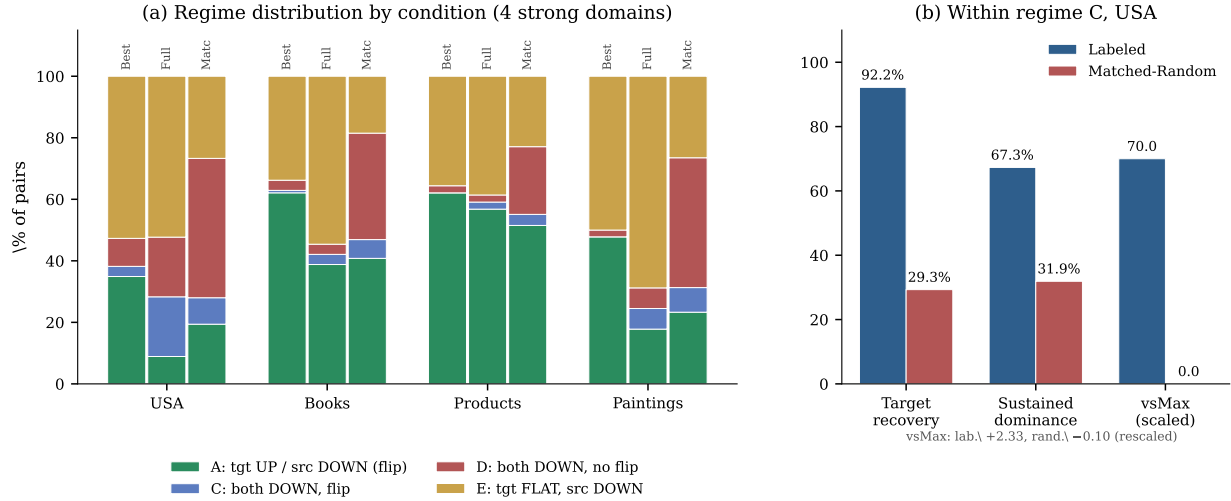


Figure 3. C1 evidence: regime decomposition and within-regime C signals. (a) Stacked-bar regime distributions (A/C/D/E) for the per-pair best variant, the full labeled ($M=20$), and matched random across the 4 strong domains. The best variant concentrates in regime A; random concentrates in regime D. (b) Within regime C in USA, labeled vs. random on three sub-regime signals: Target-Recovery rate (92.2 vs 29.3%), sustained dominance (tgt_win_pct 0.673 vs 0.319), and vsMax (+2.33 vs -0.10). Numbers in Appendix L.

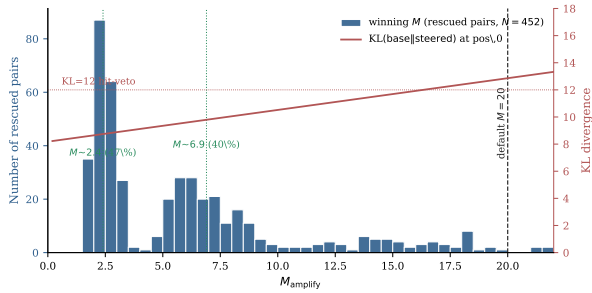


Figure 4. Bimodal winning-M distribution. Histogram of M_{amplify} values rescuing missed pairs in adaptive search ($N=452$), with $KL(\text{baseline}||\text{steered})$ overlay. The canonical $M=20$ default (dashed line) is to the right of both peaks (~ 2.4 and ~ 6.9). High- M rescue and top- k rescue are both null (Appendix G).

4.3%; see Appendix K for the per-layer profile). *Within-domain pair-level prediction is null in USA* (Q1 27.8% vs Q4 24.3%, no monotone trend; the within-USA scaffold range [0.38, 0.73] is too narrow to discriminate pairs) and only weak in smaller domains (Products residualized $r = +0.29$, $N=132$). We report this honestly in Section 6.3.

Falsification verdict on C2. Three of four sub-claims (C2.1, C2.2, C2.3) are supported with directional thresholds met in 4–5 domains. The cross-domain ordinal prediction in C2.4 is supported ($\rho=1.0$, $N=4$); the *within-domain pair-level* prediction in C2.4 is null in USA and weak elsewhere, reported as a sub-claim negative.

6.3. Where the method fails

We commit two named negative results to the main body, in line with the workshop’s call for rigorous negative findings.

Negative result 1: Sounds. On 6 entities, 12/30 non-identity pairs share the answer “brown”. Random hits (3.3%) come predominantly from these shared-answer pairs; matched-control vsMax exceeds labeled vsMax by 0.14 logits. *Sounds fails Condition 2 of Definition 2.6*. We report this as a domain-level negative and exclude Sounds from cross-domain inference for C1; we also identify three structural causes (small N , shared-answer artefacts, $K=6$ thin logit margins; Appendix N).

Negative result 2: Within-USA scaffold prediction. The scaffold metric is a perfect cross-domain ordinal predictor of Hit% but a null within-domain pair-level predictor in USA, the most powered dataset (Q1–Q4 hit rates 27.8/24.7/24.3/24.3%; $|r| < 0.10$ for all variants; $N=2,450$). The within-USA scaffold range is too narrow to discriminate pairs. *The scaffold story is a between-domain story; we explicitly do not claim within-domain predictive utility.*

7. Limitations and Threats to Validity

The workshop call is explicit that limitations must be load-bearing. We enumerate honestly.

1. No grouping-method baseline. We deliberately do not compare against geometric clustering (cosine, layer-adjacency, k -means). Any grouping derived from a behavior-

grounded signature beats a geometric grouping on behavior-grounded metrics by construction (structurally biased in our favor), and there is no canonical feature-clustering baseline in the SAE/CLT literature. We instead validate via intervention against matched random controls (Definition 2.5), which is method-agnostic and applicable to any future grouping. We encourage others to run our harness on competing groupings.

2. Attention-circuit blindspot. Attribution graphs freeze attention; swaps run free. The asymmetry plausibly explains part of the 33–79% miss rate in the strong domains (Ameisen et al., 2025). Attention-aware variants are listed as future work (Section 8).

3. Sounds dataset structural issues. Six entities, 12/30 pairs share answers, $K=6$. Sounds fails Condition 2 of Definition 2.6 (Section 6.3); excluded from cross-domain inference for C1.

4. Substring matching and multi-token heuristics. The concept-to-supernode matcher allows reverse containment (e.g., *is matching mississippi*), affecting 28–45% of entity-field combinations. Multi-word answers resolve to first subword; both can miss or false-match. Impact on swap performance is unmeasured.

5. Threshold sensitivity is unmeasured. Decision-rule thresholds were iteratively refined on pilot circuits; no formal ± 10 –20% sensitivity sweep. Thresholds are treated as configurable defaults.

6. Cross-prompt robustness is observational. $N=1,607$ pairs but observational; we do not run interventions on the overlap claim itself; the layer-gradient evidence for C2.3 is correlational.

7. Within-domain pair-level scaffold prediction is null in USA. Reported as a sub-claim negative (Section 6.3); the scaffold story is between-domain only.

8. Three levels of interpretive claim; single model and language. Per Geiger et al. (2025), we claim Levels 1–2 only. Level 3 (full mechanistic identification) is disclaimed. All experiments use Gemma-2-2B-it with CLT-HP, English single-hop prompts; cross-architecture, cross-dictionary, and multilingual generalization are open.

8. Discussion and Conclusion

Methodological lessons. Three findings generalize beyond probe prompting. (i) *Match random controls structurally.* Without matching feature count, per-layer histogram, and concept-exclusion, the labeled–random gap

is misstated; random controls produce *more* suppression than labeled in 3/5 domains. (ii) *Default $M=20$ does not transfer.* Over-steering in the multi-feature CLT setting; winning M is bimodal at $\sim 2.4/\sim 6.9$. Recommend a coarse M sweep as standard practice. (iii) *Decompose by semantic field role.* The less-is-more effect (14–33 pp gain from removing input-field features) shows that input features encode what the model *reads*, not what it should *produce*.

What the labels mean. Supernode labels are behavioral abstractions; they pass an operational test against matched controls but do not constitute mechanistic identifications of latent variables (Level 3 of Geiger et al., 2025).

Future work. (a) Attention-aware variants; (b) cross-architecture replication on Gemma-2-9B and Llama-3; (c) multilingual extension; (d) within-domain pair-level scaffold predictor. All artifacts will be de-anonymized at camera ready.

Conclusion. Probe prompting demonstrates that circuit-level interpretability of attribution graphs can be partially automated using a transparent rule-based pipeline validated by intervention against matched random controls. The two-condition definition of an operationally useful label gives the harness a sharp falsification criterion; its substantive findings (less-is-more, over-steering, scaffold) inform best practice in the steering literature.

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A. Reproducibility Statement

To support the workshop’s emphasis on reproducibility and code/data access, we release the following anonymously for the review period.

- **Code.** Anonymous repository at <https://anonymous.4open.science/r/attribution-graph-probing-anon> (commit hash and license in README.md).
- **Datasets.** The full 42,184-run swap dataset (graphs, prompts, activations, supernode groupings, swap dumps for all five domains) at the same anonymized URL, with a per-domain manifest in Appendix O.
- **Interactive demo.** An anonymized HuggingFace Space at <https://anonymous.4open.science/r/attribution-graph-probing-demo-anon> that accepts an attribution graph and returns concept-aligned supernodes with the matched-control diagnostics.
- **Per-domain CLI examples.** Appendix O gives end-to-end commands for the USA, Books, Products, Paintings, Sounds pipelines, including the exact τ , M , and seed used for each table and figure.
- **Determinism.** The pipeline is fully deterministic (sha256-seeded random controls; fixed temperature/seed/penalty for generation; no random initialization); per-feature checkpoints permit resumption from partial runs.
- **Hardware.** A single L4 GPU is sufficient to reproduce any single domain in 14–24 minutes (graph generation + activation measurement + grouping + subgraph), excluding the swap sweep; the full swap sweep is documented per-domain in Appendix O.

We commit to de-anonymizing the URLs at camera ready and to maintaining the artifacts indefinitely.

Appendix: How to Read

Appendix material is provided for completeness; main-paper claims are self-contained in the 8-page body. The appendices fall into five groups: (Repro.) Reproducibility Statement; (A–C) pipeline reference and metric justifications; (D–H) per-domain quantitative tables and metric variants; (I–K) cross-prompt and scaffold analyses; (L–M) case studies and the Sounds/Paintings deep dive; (N–O) reproducibility manifest and glossary.

B. Pipeline Reference

B.1. Pseudocode

```

Input: attribution_graph G_p, seed_prompt p, target_logit y
1. V_tau := select_nodes_by_cumulative_influence(G_p, tau=0.95)
2. Concepts := llm_generate_concepts(p, K)
3. Probes := synthesize_probes(Concepts, template_constraints)
4. for f in V_tau:
    for q in Probes:
        A[f, q] := measure_activation(f, q)           # Neuronpedia API or local
5. CPAS := aggregate_cross_prompt(A)                 # Sec 3.2
6. for f in V_tau:
    role[f] := classify(CPAS[f], thresholds)         # Sec 3.4 strict-priority rules
    name[f] := assign_name(f, A, blacklist, target_token_rules)
7. Supernodes := group_by(role, name) with stability >= 0.6
8. Subgraph := pin_nodes(Supernodes, embeddings, y)
9. for (e_A, e_B) in pairs:
    I_lab := build_swap(S(e_A), S(e_B), M_a=-2, M_b=20)
    I_rand := matched_random_control(I_lab)           # Def 5
    run_swap(I_lab); run_swap(I_rand)                 # freeze_attention=false
10. Adaptive_M_search on missed pairs                # Sec 3.7
    
```

B.2. CPAS metrics

- `peak_consistency_main`: fraction of probes whose peak token matches the modal peak.
- `n_distinct_peaks`: number of distinct peak tokens across active probes.

- `share_F` / `conf_F`: fraction of active probes whose peak is on a functional token / classifier confidence on that share.
- `func_vs_sem_pct`: $100 \cdot (\max \text{func} - \max \text{sem}) / \max \text{overall}$.
- `sparsity_median`: median of $(\text{peak} - \text{mean}) / \text{peak}$ across active probes.
- `conf_S`: $1 - \text{share}_F$.

B.3. Decision-rule thresholds (frozen pre-analysis)

SEM-DICT `peak_consistency` ≥ 0.80 AND `n_distinct_peaks` ≤ 1 . SAY-X `func_vs_sem_pct` ≥ 50 AND `conf_F` ≥ 0.90 AND `layer` ≥ 7 . REL `sparsity_median` < 0.45 . SEM-CONC `layer` ≤ 3 OR `conf_S` ≥ 0.50 OR `func_vs_sem_pct` < 50 . Otherwise REVIEW (excluded).

C. Functional Vocabulary and Target-Token Mapping

English functional tokens (87 total). **Copulas:** is, was, are, were, be, been, being, am, 's, 're, 'm. **Articles:** the, a, an, this, that, these, those. **Prepositions:** of, in, on, at, to, for, with, by, from, as, into, onto, upon, about, above, below, between, among, through, during, before, after. **Conjunctions:** and, or, but, nor, so, yet, for. **Relative pronouns:** that, which, who, whom, whose, where, when, why, how. **Auxiliaries:** do, does, did, have, has, had, will, would, shall, should, can, could, may, might, must.

Directionality. *Forward:* copulas, auxiliaries, articles, most prepositions. *Backward:* of, possessive 's. *Bidirectional:* conjunctions, punctuation. Window: 7 tokens. Multi-token entities resolve to first subword.

Documented French failure mode. Applying the English pipeline to “le contraire de ‘petit’ est” with English probes produced incoherent activations and no valid supernode grouping. After adding French probes and expanding the blacklist (mixed FR/EN bridging tokens), grouping became reliable; we report this as a known cross-lingual limitation.

D. Why Gap-Closure Was De-Emphasized

Gap-closure is regime-dependent and confounded by baseline gap magnitude:

Regime	GapCl (lab)	GapCl (rand)	vsMax (lab)	vsMax (rand)
A	0.88	0.76	+4.28	−0.19
C	2.73	1.59	+2.33	−0.10
D	13.62	6.46	+4.11	−0.07

Regime D shows the *highest* gap-closure despite being the weakest evidence regime—both tokens start far apart and both get disrupted, leaving room for “closure” without the target ever winning. Cross-domain, GapCl ranks Books ~ 0 but vsMax ranks Books strongest (+5.98). GapCl is therefore not used as a primary discriminator.

E. Per-Domain Full-Scale Labeled vs Random

E.1. Headline (full labeled, all fields, $M=20$)

Domain	Cond	N	Hit%	Sup%	vsMax	RkGrp	MedRk	Flip%
USA	lab.	2,450	24.7	92.8	+2.86	1.72	5	98.2
	rand	7,350	0.1	83.4	−2.31	9.00	566	69.2
Books	lab.	240	3.8	69.2	+5.98	1.03	17	96.7
	rand	720	0.3	75.0	−0.15	2.43	283	79.6
Prods	lab.	132	15.2	65.2	+3.46	1.20	26	97.0
	rand	396	0.3	87.1	+0.23	2.25	354	75.3
Paint	lab.	90	4.4	34.4	+1.55	1.31	70	97.8
	rand	270	0.0	74.1	−0.03	1.96	196	88.9
Sound	lab.	30	0.0	100	+3.28	1.00	21	60.0
	rand	90	12.2	80.0	+3.14	1.08	24	53.3

E.2. Specificity strength

Books +6.13 (Strong); USA +5.17 (Strong); Products +3.23 (Moderate); Paintings +1.58 (Weak); Sounds +0.14 (Negligible).

F. Field-Additivity Detail

F.1. Per-domain 7-variant tables

The six tables below give Hit%, Sup%, GapCl, MedRk, vsMax, vsTopK, RkGrp for all 7 field variants per domain. Bold marks the best-Hit% subset.

USA (state, capital, city; 2,450 pairs/variant).

Subset	Hit%	vsMax	RkGrp	MedRk
city	2.9	-1.26	6.92	190
state	17.0	+2.38	2.72	14
capital	14.2	+0.63	3.69	11
state+capital	38.8	+4.00	1.47	3
state+city	11.1	+1.71	2.31	28
capital+city	10.2	+0.58	4.16	30
all 3	24.7	+2.86	1.72	5

Books (character, book, author; 240 pairs/variant).

Subset	Hit%	vsMax	RkGrp	MedRk
character	4.2	+1.87	1.77	125
book	37.1	+6.69	1.20	2
author	14.6	+4.71	1.33	12
book+author	37.1	+7.76	1.02	2
character+book	5.0	+5.22	1.23	24
character+author	3.8	+3.45	1.30	76
all 3	3.8	+5.97	1.03	17

Products (product, company, founder; 132 pairs/variant).

Subset	Hit%	vsMax	RkGrp	MedRk
product	0.0	+1.90	1.61	137
company	5.3	+2.62	1.39	128
founder	8.3	+2.08	1.27	18
company+founder	24.2	+3.06	1.27	18
product+company	1.5	+2.78	1.31	93
product+founder	2.3	+2.54	1.23	48
all 3	15.2	+3.47	1.20	26

Paintings (painting, painter, first_name; 90 pairs/variant).

Subset	Hit%	vsMax	RkGrp	MedRk
painting	3.3	+1.50	1.44	71
painter	1.1	+1.68	1.30	66
first_name	6.7	+1.46	1.41	90
painter+first_name	1.1	+1.69	1.30	65
all 3	3.3	+1.55	1.32	66

Sounds (sound, animal, color; 30 pairs/variant).

Subset	Hit%	vsMax	RkGrp	MedRk
sound	20.0	+5.49	1.00	5
animal	12.5	+3.49	1.21	6
color	0.0	+3.09	1.07	48
sound+animal	20.0	+4.69	1.07	5
all 3	0.0	+3.40	1.00	20

F.2. Cross-domain single-field role aggregates

Role	Hit%	Sup%	MedRk	vsMax	RkGrp
Input	6.1	63.2	106	+1.90	2.55
Intermediate	14.6	79.1	43	+3.37	1.56
Answer	8.8	89.5	36	+2.39	1.75

G. Adaptive M-Search

G.1. Two-phase protocol

Phase 1 (coarse, geometric): $M \in \{0.1, 0.25, 0.6, 1.5, 4.0, 10.0, 20.0\}$ probed in order; stop at first Hit. **Phase 2 (KL-binary):** compute $KL(M)$ at the Phase-1 grid; identify the steepest KL transition; binary-refine M in that interval for 6 steps. Accept hit at any refinement point.

G.2. Per-domain rescue

Cond.	Domain	Eligible	New hits	Hit-rate	Before	After	Δ
Lab.	USA	870	329	37.8	24.7	38.2	+13.5
Lab.	Books	85	12	14.1	3.3	9.0	+5.7
Lab.	Paint	103	7	6.8	4.0	9.7	+5.6
Lab.	Sound	23	5	21.7	0.0	16.7	+16.7
Lab.	Prods	128	4	3.1	13.2	15.5	+2.3
FA	USA	553	78	14.1	64.2	67.3	+3.2
FA	Books	73	9	12.3	59.5	63.8	+4.3
FA	Paint	96	6	6.3	16.9	21.8	+4.8
FA	Prods	124	2	1.6	26.4	27.6	+1.1
FA	Sound	21	0	0.0	20.0	20.0	+0.0
Total	—	2,076	452	—	—	—	—

G.3. Winning- M histogram

93% of hits found in Phase 1; bimodal at $M \sim 2.4$ (47%) and $M \sim 6.9$ (40%); long tail to $M \geq 10$ (13%).

G.4. High- M negative result

$M \in \{50, 100, 200\}$ on near-miss pairs across 3 domains: 80 runs, 0 new hits; existing hits at $M=20$ degraded monotonically with increasing M .

G.5. Top- k negative result

$k \in \{1, 3, 5, 10\}$ by graph influence at $M \in [20, 200]$ on Products: 96 runs, 0 hits. Only the full feature set ($k=67$) achieves hits. Graph influence correlates negatively with stored activation ($\rho = -0.362$, $p=0.003$).

G.6. Five failure modes

(1) *Severe overshoot* (rescuable by M reduction; 88% rescue rate; 37/42 oklahoma_tulsa pairs at $M=5$). (2) *Field interference* (rescuable by mid+answer subset; Appendix F). (3) *Feature interaction noise* (garbage at high M despite positive rank improvement; indiana→arkansas: tonode at $M=20$ → Little Rock at $M=5$). (4) *Signal collapse* (vsMax collapses below pair-specific M ; vermont→kansas: vsMax 8.3 at $M=20$ → 0.3 at $M=5$). (5) *Feature specificity* (correct context, wrong answer; Hutchinson not Topeka; not fixable by M).

H. Position-0 vs Best-Over-Trajectory Metrics

We report both variants because they diverge informatively.

vsMax > 0 rate (best variant).

Domain	Lab pos0	Lab best	Rand pos0	Rand best
USA	52.9	91.0	2.8	13.1
Books	92.5	98.8	4.6	45.8
Products	70.5	75.0	15.2	51.5
Paint	61.1	81.1	14.4	47.8
Sounds	50.0	90.0	63.3	93.3

Position-0 vsMax > 0 is a *strict necessary condition* for output success in Books and Products (zero hits when pos0 vsMax ≤ 0); USA tolerates trajectory recovery (28.7% of USA hits have pos0 vsMax ≤ 0); Sounds reveals *no signal at pos0* (random exceeds labeled).

Mean vsMax (Books illustration). Labeled pos0 +7.29 vs random pos0 −5.80 = gap +13.09; best-over-trajectory gap is only +8.01. Position-0 nearly doubles the effect size.

The labeled-vs-best-variant pos0 split (USA). Labeled pos0 vsMax > 0 rate: 27.2% (median pos0 vsMax −3.88) vs best variant 52.9% (median +0.31). Position-0 evidence for the less-is-more effect: input-field features cause initial disruption.

I. KL Divergence as a Steering Diagnostic

Linearity with M . Per-pair KL(M) is highly linear ($R^2 > 0.93$ for 8/10 USA pairs; range 0.85–1.00). Mean KL by M : $M=5 \rightarrow 9.33$; $M=10 \rightarrow 11.10$; $M=20 \rightarrow 13.03$.

KL ≥ 12 as a hit-veto. Across 30 USA observations, no hit occurs at KL ≥ 12. As a binary classifier on {KL < 12}: recall 100% (7/7), precision 37% (7/19; 12 false positives split among feature-specificity (3), evaluator gaps (3), signal collapse (3), near-threshold (3)).

KL slope vs intervention structure. $r(\text{slope}, \text{ablate_count}) = +0.68$; $r(\text{intercept}, \text{ablate_count}) = -0.63$. More source features ablated → steeper KL increase per unit M . High-slope pairs are most sensitive to M and most likely to benefit from M -reduction.

Use in adaptive search. (i) Run at $M=20$ and measure KL at pos 0; (ii) if KL ≥ 12, fit KL(M) from ≥ 1 extra point and compute $M_{\text{crit}} = (12 - b)/a$; (iii) re-run at $M = \lfloor 0.8 \cdot M_{\text{crit}} \rfloor$; (iv) if intercept $b \geq 12$, flag as intrinsically disruptive.

J. Cross-Prompt Robustness at Scale

1,607 intra-domain pairs, 5 domains, 103 entities. 5,000 bootstrap resamples; 2,000 permutations per pool size.

Jaccard (95% CI). USA 0.465 [0.462, 0.468], $N=1,225$; Books 0.308 [0.302, 0.315], $N=210$; Products 0.364 [0.356, 0.374], $N=91$; Paintings 0.286 [0.279, 0.292], $N=66$; Sounds 0.621 [0.597, 0.648], $N=15$. All $p < 0.001$ at every tested pool size.

Activation stability. USA 0.947, Books 0.908, Products 0.903, Paintings 0.916, Sounds 0.944. Peak-token agreement 85–98%; peak-type agreement 93–99%.

Layer-binned Jaccard (early/mid/late) and early/late ratio. USA 0.543/0.440/0.293 (1.85×); Books 0.347/0.340/0.184 (1.89×); Products 0.496/0.308/0.164 (3.02×); Paintings 0.302/0.311/0.212 (1.43×); Sounds 0.684/0.544/0.440 (1.56×).

Supernode consistency. Same/regrouped/inconsistent: USA 76.6/16.2/7.3; Books 47.5/12.8/39.7; Products 65.2/16.5/18.4; Paintings 71.0/9.9/19.1; Sounds 85.6/8.9/5.6.

Caveats. (i) Original Dallas/Oakland pair is a ~93rd-percentile USA outlier; population means are 10–15 pp lower than originally reported. (ii) Books inconsistency rate likely reflects keyword-detection limits on literary names, not genuine instability. (iii) Paintings shows a weak/inverted layer gradient that is unexplained. (iv) Sounds has high overlap but 0% hit, confirming overlap is necessary but not sufficient.

K. Graph Compatibility (Scaffold) Analysis

Definitions. *Scaffold*: same `feature_key` and same supernode in both graphs (structural primitives like copulas/prepositions/task operators). *Regrouped*: same feature, different supernode (entity-appropriate re-assignment such as Say (Austin) $\rightarrow$ Say (Sacramento)). *Entity-only*: present in one graph only.

Per-domain scaffold influence and per-layer profile. Cross-domain ranking: USA 53.0%, Products 42.2%, Paintings 35.9%, Books 25.3%. Late-layer ($\geq L_{16}$) scaffold: USA 16.3%, Products 10.3%, Paintings 9.2%, Books 4.3%. Both rank-perfect ordinal predictors of full-labeled Hit%.

Within-domain. USA: all $|r| < 0.10$, Q1 27.8% vs Q4 24.3% (no monotone trend). Products: target-residualized $r = +0.29$, $N=132$, concentrated in L_{11-16} and L_{20-21} . Books/Paintings: positive but $N < 10$ hits, fragile.

L. Logit-Shift Regime Taxonomy

Definitions (position 0). **A**: target UP, source DOWN, flip. **C**: both DOWN, flip (differential disruption with target overtake). **D**: both DOWN, no flip (generic disruption). **E**: target FLAT, source DOWN (pure suppression).

Regime A prevalence (best variant / full labeled / random). USA 34.9/8.9/19.4; Books 62.1/38.8/40.8; Products 62.1/56.8/51.5; Paintings 47.8/17.8/23.3.

Regime D prevalence. USA 9.1/19.4/45.3; Books 3.3/3.3/34.6; Products 2.3/2.3/22.0; Paintings 2.2/6.7/42.2.

Within regime C (USA).

Signal	Labeled	Random
Target recovery rate	92.2	29.3
Sustained dominance (tgt_win_pct)	0.673	0.319
vsMax	+2.33	-0.10
Hit%	22.5	0.6

M. Individual Case Studies Grid

M.1. Domain distribution (best field-add variant)

Domain	N	Hit%	RkGrp=1	A	B	C	D
USA	2,450	38.8	91.8	35.4	55.2	3.3	6.0
Books	240	37.1	98.8	36.3	60.4	0.8	2.5
Prods	132	24.2	76.5	24.2	72.7	0.0	3.0
Paint	90	1.1	81.1	1.1	96.7	0.0	2.2
Sound	30	20.0	93.3	0.0	50.0	20.0	30.0

M.2. Six success cases

- **Colorado Springs** $\rightarrow$ **Detroit** (USA, state+capital): vsMax +17.5, target rank 342 $\rightarrow$ 1; "...Lansing..." Textbook regime A.
- **Holden Caulfield** $\rightarrow$ **Katniss Everdeen** (Books, book+author): vsMax +21.79 (largest in dataset), 11 ablate / 63 amplify; "...Suzanne Collins..."
- **Nike** $\rightarrow$ **Tesla** (Products, company+founder): vsMax +12.25, target rank 5,933 $\rightarrow$ 1; "Elon Musk..." (steered first-token $p=0.82$).
- **La Grande Jatte** $\rightarrow$ **Water Lilies** (Paintings, painter+first_name): vsMax +9.375, target baseline rank 3 (the only painting hit; 1/90).
- **Bark** $\rightarrow$ **Hoot** (Sounds, sound+animal): mechanical artifact—both share answer "brown"; not evidence of label quality.

- **Minnesota→Florida** (USA, all-3-fields labeled): vsMax +16.06 but flip@0=False; CtrlS 16.89 (high collateral disruption); illustrates the all-fields cost.

M.3. Seven failure cases (covering five failure modes)

- **Kansas/Wichita→New Hampshire/Manchester**: total failure, vsMax −6.875, target rank 5,816, garbled “Oklahoma City...”
- **Utah/Provo→Iowa/Cedar Rapids**: vsMax +14.0, RkGrp 1, but garbled output (rank-hit misalignment).
- **Ohio/Cleveland→Oklahoma/Tulsa**: regime D (no flip@0 despite vsMax +13.31); autoregressive decoding never recovers.
- **Scout Finch→Huckleberry Finn**: 176 amplify features (highest in Books), garbled (less-is-more violation; substring confound on `finn`).
- **Windows→Oculus** (Products): suppressed source, but produces “Rift” (product), not “Palmer Luckey”—“company” supernodes encode product identity.
- **Girl with a Pearl Earring→Guernica**: 195 amplify / 88 ablate fail to dislodge Vermeer; target “Pablo” rank 1,382.
- **Hiss→Hoot** (Sounds): vsMax −0.5; intervention nudges target “brown” to rank 2 while “black” takes rank 1 (tiny answer-space).

M.4. Suppression-only zone (source absent, no target)

USA 59.2%, Books 48.3%, Products 46.2%, Paintings 68.9%, Sounds 66.7%. The single largest outcome category in every domain: suppression is generic; targeting requires specific features.

M.5. Threats to validity for case studies

Selection bias toward extremes; tier unavailability for the best-FA variant at initial analysis; sounds shared-answer artefacts (12/30 pairs); single-example interpretations should not be over-generalized; explanations are post-hoc and constitute hypotheses, not conclusions.

N. Sounds and Paintings Deep Dive

Why Sounds fails Condition 2. Three structural causes converge: (i) $N=6$ entities; (ii) 12/30 non-identity pairs share the answer “brown”, so any source-suppression in those pairs incidentally counts as a target-hit for random; (iii) $K=6$ colors with near-uniform ~ 0.15 probability mass make the logit margin too thin to overshoot precisely. The vsMax gap is +0.14 (Negligible); position-0 vsMax actually *favors* random (+0.30 vs labeled +0.68 in median terms; Appendix H).

Why Paintings is weak. The `first_name` answer field is fundamentally coarse: Claude, Pablo, Leonardo are common tokens with many non-painting associations. The `painter` supernode subsumes `first_name` (since “Monet” contains information about “Claude”), making the two fields effectively identical for matching. With 17% error-node rates (highest of any domain), much of the circuit is invisible. 89/90 pairs fail at the `painter+first_name` variant; the only hit (Grande Jatte → Water Lilies) requires baseline rank 3.

Recommendation. We exclude Sounds from cross-domain inference for C1 and treat Paintings as a Weak-strength supporting domain rather than a confirmatory one. The four-domain rank-perfect scaffold prediction (Appendix K) holds without Sounds.

O. Reproducibility and Tool Access

Code, data, demo (all anonymized for review). Repository: <https://anonymous.4open.science/r/attribution-graph-probing-anon> (commit hash and license in README.md). Demo: <https://anonymous.4open.science/r/attribution-graph-probing-demo-anon>. License: GNU GPL v3.

Per-domain CLI (USA example).

```
# 1. Generate attribution graph (Neuronpedia API)
python scripts/00_neuronpedia_graph_generation.py \
  --model gemma-2-2b-it \
  --prompt "The capital of the state containing Dallas is" \
  --target "Austin" \
  --node-threshold 0.8 --edge-threshold 0.85 --max-nodes 5000 \
  --output_dir output/usa/dallas_austin/

# 2. Select features (interactive UI)
streamlit run eda/threshold_selection.py --tau 0.95

# 3. Probes + activations + grouping
python scripts/01_probe_prompts.py --graph output/usa/dallas_austin/graph.json \
  --output_csv output/usa/dallas_austin/activations.csv
python scripts/02_node_grouping.py --input output/usa/dallas_austin/activations.csv \
  --graph output/usa/dallas_austin/graph.json \
  --output output/usa/dallas_austin/grouped.csv

# 4. Run swap protocol with matched random control + adaptive M-search
python scripts/03_swap_full.py --domain usa \
  --m_ablate -2 --m_amplify 20 --freeze-attention false \
  --random-replicates 3 --random-seed 42 \
  --field-additivity true --m-search adaptive \
  --output_dir output/usa_states_batch/
```

Runtime profile. Per single-circuit pipeline (graph + activation + grouping + subgraph): 14–24 minutes on a single L4 GPU. Full per-domain swap sweeps: USA 17h, Books 1.6h, Products 0.9h, Paintings 0.6h, Sounds 0.2h on 8 H100 GPUs (matched-control replicates parallelized).

Artifact release manifest. For each domain: (a) attribution graphs (Neuronpedia JSON); (b) probe-prompt files; (c) activation matrices; (d) supernode groupings; (e) swap dumps with per-pair Hit/vsMax/RkGrp/Sup/Flip/CtrlS/KL trajectories; (f) field-additivity per-variant tables; (g) adaptive- M logs (Phase 1 & Phase 2); (h) random-control sha256 seeds.

Determinism and resume. Pipeline is fully deterministic (sha256-seeded controls; $T=0.3$ / $n=10$ / freq-penalty 2.0 / seed 42; no random initialization). Per-feature checkpoints permit resume from partial runs.

P. Glossary and Notation

CLT: Cross-Layer Transcoder. **CPAS:** Cross-Prompt Activation Signature. τ : cumulative-influence threshold for feature selection (default 0.95). M_{ablate} , M_{amplify} : multipliers for source/target features (defaults -2 , 20). **vsMax:** $\text{best}_t(\ell_{e_B}(t) - \max_{e' \neq e_B} \ell_{e'}(t))$. **RkGrp:** best target rank within domain answer set. **Target Recovery:** target logit beats its own unsteered baseline at any trajectory position. **Regime A/C/D/E:** position-0 logit-shift regimes (Appendix L). **Scaffold:** features shared between source/target graphs with identical supernode assignment. **Scaffold influence:** fraction of total graph influence carried by scaffold features. **KL:** $\text{KL}(\text{baseline} \parallel \text{steered})$ at position 0.